Decentralised Single-Index Bandits

Learning an unknown reward function without a server: theorems and findings

Presented by:

Harshitha (23b1071) Dept. of CSE
Nischal (23b1024) Dept. of CSE

Course: CS6007: Multi-Agent Machine Learning

Instructor: Prof. Avishek Ghosh

IIT Bombay

25 October 2026

Problem statement

Setting

N agents on a graph $\mathcal{G}$, **no server**. Each round agent i sees K contexts $x \in \mathbb{R}^d$, pulls one, observes $y = f(\langle x, \theta^* \rangle) + \eta$. Unknown, shared θ^* and f (f possibly non-monotone). Goal: small network regret $R_T(N)$, little communication.

Key difficulties

- **Two unknowns:** the direction needs pooled data; the shape of f needs per-bin counts.
- **No coordinator:** totals are built over the graph, $\approx$ diameter rounds late.
- **Stale or averaged counts** break the UCB confidence argument.
- **Communication:** sharing tables every round costs $\Theta(|E|N_b T)$ scalars.

What prior work misses

Single-index bandits are single-agent; decentralised bandits assume finite arms, linear or kernel models. No decentralised single-index bandit, and no graph-dependent lower bound for this class.

Literature survey

Work	What it does	Use / gap
Dey, Bhore & Ghosh (2026)	one agent, non-monotone f; optimal $\tilde{\Theta}(T^{2/3})$	our base algorithm; no network
Kang et al. (ICLR 2026)	ESTOR (monotone f), GSTOR ($T^{3/4}$)	baselines; one agent
Wang et al. (ICLR 2020)	bandits *with a server*; event-triggered sync	our trigger; needs a server
Landgren (2016); DDUCB (NeurIPS 2019)	decentralised K-armed; consensus, gossip	mixing baselines; finite arms
DLUCB (AAAI 2021)	decentralised *linear* bandits; delayed, consistent statistics	our epoch idea; linear model
Coop-KernelUCB (ICML 2020)	kernels, message passing; graph lower bound *open*	relaying; costly in d

Gap: an unknown index *and* an unknown non-monotone link, learned by agents that talk only to neighbours.

Contributions ⋆

1. **New problem:** the first decentralised single-index bandit. Agents share θ^* and f and talk only over a graph.
2. **Lower bounds:** no serverless algorithm beats $\Omega((NT)^{2/3})$ (Thm 5). On rings and paths it must also pay $\Omega(N \min(T, D)^{1/3})$ (Thm 7, light cones).
3. **Upper bounds:** the graph costs only an *additive, horizon-free* term: $\tilde{O}\big((dNT)^{2/3} + N\Lambda K\big)$ (Thm 22). A provable relay with $\tilde{O}\big((dNT)^{2/3} + KF'\big)$, where F' is the farness (Thm 27), beats the relay we first ran.
4. **Communication and robustness:** logarithmic in T for a fixed grid (Thm 29), $O(N_b + N)$ memory per agent. Directed, delayed and time-varying graphs change only the delivery time Λ, and our bound $B(N-1)$ is tight.
5. **Experiments:** 16 studies with paired trials and 95% CIs. Prior papers' algorithms (DLUCB-, ESTOR-, GSTOR- and kernel-style) on our problem. About 12k lines of Python, all figures reproducible.

Recap: single-index bandits

Model

Each round an agent sees K arms $x_{t,a} \in \mathbb{R}^d$, pulls one, and observes

$$\boxed{y_t = f(\langle x_t, \theta_* \rangle) + \eta_t}$$

θ_* and f unknown, f possibly non-monotone.

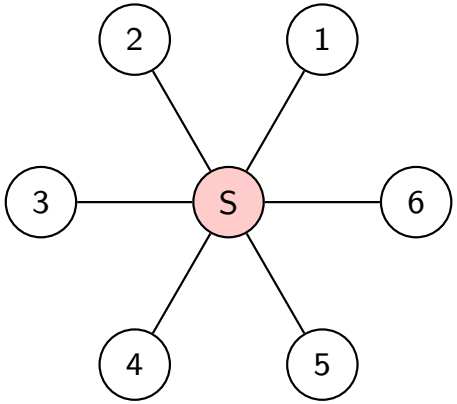
ZoomSIB-UCB (Dey, Bhore & Ghosh 2026): optimal $\tilde{\Theta}(T^{2/3})$

- **Phase 1:** explore; estimate θ_* with the Stein estimator $\hat{v} = \frac{1}{T_0} \sum_t y_t S(x_t) \propto \theta_*$, a *sample mean*.
- **Phase 2:** project $z = x^\top \hat{\theta}$, cut the line into bins, run *sleeping* UCB over the bins available each round. State: (n_j, S_j) per bin.

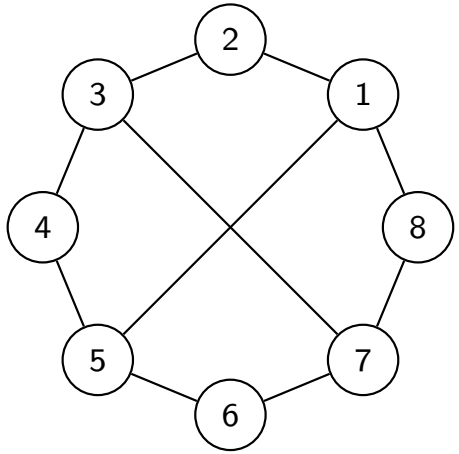
Our question

N agents share θ_* and f but **no server**: they talk only to graph neighbours.

With a server vs without a server



Server (star)
one coordinator sees all statistics



Serverless (graph G)
diameter D, spectral gap $1 - \lambda$

Why serverless?

No trusted coordinator, or no backbone: sensor networks, robot/drone swarms, vehicular networks, peer-to-peer recommendation, consortia that do not share raw data.

Network regret $R_T(N) = \sum_{i,t} \left[f(\langle x^*_{i,t}, \theta_* \rangle) - f(\langle x_{i,t}, \theta_* \rangle) \right]$

The algorithm: Dec-ZoomSIB (published-only relay)

Every agent i runs the same code

Phase 1: learn the direction

1: **for** $t \le T_0$: pull a random arm; $b_i \leftarrow b_i + \varphi_\tau\big(y\, S(x)\big)$ ▷ truncated Stein sum

2: flood b_i; wait Λ rounds ▷ every agent now holds all N sums

3: $\hat{\theta} \leftarrow \sum_k b_k / \|\sum_k b_k\|_1$ ▷ identical everywhere; equals the server's

Phase 2: learn f with sleeping UCB on bins of width Δ along $\langle x, \hat{\theta}\rangle$

4: **for** each round t:

5: $(n_j, S_j) \leftarrow \sum_k$ freshest *published* table of k ▷ own table too, once published

6: pull an arm in the available bin with the largest $S_j/n_j + c_\delta/\sqrt{n_j}$

7: **if** own unpublished pulls in some bin j exceed $\gamma \max(n_j, 1)$: flood own table

Why this design

- Phase 1 is **exact**: Stein sums are sufficient statistics, so flooding them gives the server's $\hat{\theta}$.
- Phase 2 relays **exact, sparse** tables, published only when a bin grows by a factor γ.
- Acting only on *published* data is the key to the proof (Thm 27).

Theory: setting of the proofs

Assumptions (DBG's, with one made explicit)

- Contexts i.i.d., coordinates κ-sub-Gaussian, $\mathbb{E}[S_j(X)^2] \le M$; $|f| \le L_f$, $|f'| \le L_{f'}$ on $|z| \le X_{\max}$; $\mu^* = \mathbb{E} f'(\langle X, \theta^* \rangle) > 0$.
- Noise i.i.d. and σ-sub-Gaussian. DBG's Stein lemma already needs i.i.d. samples, and their hard instance has $\mathcal{N}(0,1)$ noise.

Communication: only one number about the graph

Delivery time Λ: a flooded message reaches every agent within Λ rounds.
Static connected graph: $\Lambda = D$ (diameter). Everything below uses the graph *only* through Λ.

Results (`docs/proofs/serverless.pdf`)

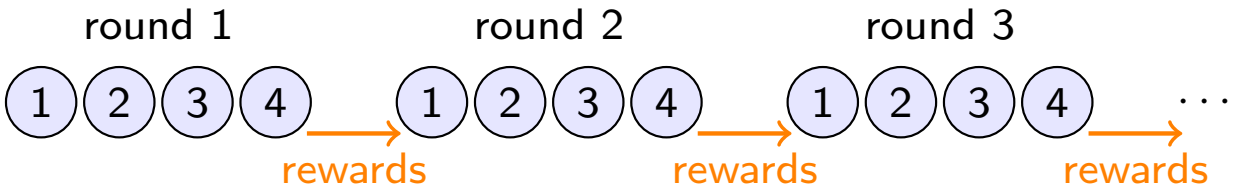
Thm 5, 7 lower bounds · Thm 17, 22, 27 regret · Thm 29 communication · Lemma 31 robustness · Thm 33, Prop 34 agreement & stopping · Thm 35 privacy

Theorem 5: no serverless algorithm beats $(NT)^{2/3}$

Theorem

For any N-agent algorithm, on any graph, even one where every agent sees all data after every round: $\sup_{\text{instances}} \mathbb{E}\, R_T(N) \geq c\,(NT)^{2/3}$.

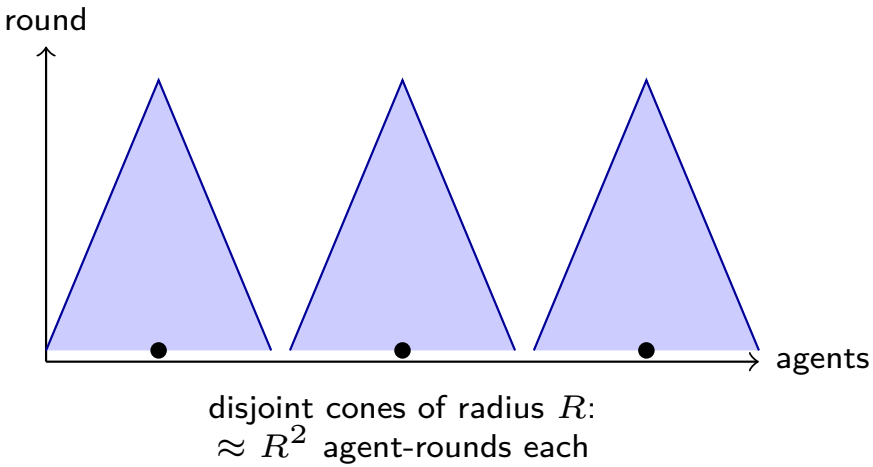
Proof idea: one agent can simulate the whole network.



- Single-agent round $u = (t-1)N + i$ plays what agent i would play in round t.
- It **withholds** round t's rewards until all N agents of round t have acted, then runs one communication step.
- Same law of the history $\Rightarrow$ network regret $=$ regret of one agent over NT rounds $\geq c(NT)^{2/3}$ (DBG Thm 5.1).

Hard instances: DBG's at horizon NT ($d = 1$, Gaussian contexts, $K \approx (NT)^{1/3} \log NT$ arms).

Theorem 7: the graph does cost regret (light cones)



Lemma 6 (light cones are closed)

Information moves one hop per round, so the actions inside a cone depend only on data inside it.

Theorem 7 and Corollary 8

Each cone acts like one agent with $n \approx R^2$ pulls, so it pays $\Omega(n^{2/3})$. One common hard instance makes all the cones pay at once. On a ring or a path:

$$\sup \mathbb{E}\, R_T(N) \;\geq\; c \max\left\{ (NT)^{2/3}, \quad N \min(T, D)^{1/3} \right\}.$$

- The second term wins when $T \lesssim \sqrt{ND}$: there, **no network on a ring can match one agent with NT pulls**.
- Longer horizons: upper graph term KF' (Thm 27) vs this $ND^{1/3}$. Linear in distance necessary? Open: a cone already learns at rate $n^{2/3}$ on its own data.

Theorem 17: regret with synchronised epochs

Synchronised epochs: a Phase 2 variant in which all agents act on the *same* delayed pooled table plus their own pulls (see backup). It is the first version we could analyse.

Theorem (with probability $\geq 1 - 3\delta$)

$$R_T(N) \leq \underbrace{2L_f N(T_0+\Lambda)}_{\text{Phase 1}} + \underbrace{4\sqrt{2}\, c_\delta \sqrt{(1+N\gamma)N_b NT}}_{\text{UCB}} + \underbrace{2L_{f'}\Delta NT}_{\text{bin width}}$$
$$+ \underbrace{4L_{f'} X_{\max}\varepsilon_0 NT}_{\text{direction error}} + \underbrace{2L_f N_b\big(2+N(\gamma+4\Lambda)\big)}_{\text{delay: the only graph term}}$$

Corollary (rate): $\Delta = (NT)^{-1/3}$, $\gamma = \gamma_0/N$, T_0 balanced

$$R_T(N) = \tilde{O}\Big((d\,NT)^{2/3} \;+\; N\Lambda\,(NT)^{1/3}\Big)$$

Matches Theorem 5 in N and T when $N\Lambda \leq (NT)^{1/3}$: **the graph costs only an additive term.**

Proof in three steps

1. *Confidence:* view = common prefix + own block $\Rightarrow$ server width, no $\sqrt{N}$.
2. *Per pull:* regret $\leq 2\times$width + bin width (works for any sleeping pattern).
3. *Stale counts:* $\leq \gamma G + 4\Lambda$ unseen pulls per agent per bin.

Theorem 22: the graph's cost does not grow with T

Observation

Theorem 17 lets *every* bin have $\approx 4N\Lambda$ unseen pulls (a factor $N_b \sim (NT)^{1/3}$). But a pull into bin j needs an arm in bin j, and there are only K arms per agent-round, so in total about $N\Lambda K$ pulls are unseen.

Theorem 22 (w.p. $\geq 1 - 4\delta$; Lemma 21: Chernoff on how often each bin is available)

$$\text{delay} \leq \underbrace{(2L_f + 4\sqrt{2}c_\delta)16N\Lambda \min(K, N_b)}_{\text{graph; no } T,\ N_b} + \underbrace{\tilde{O}(N_b)}_{\text{no graph}} \Rightarrow R_T(N) = \tilde{O}\left((dNT)^{2/3} + N\Lambda K\right)$$

epochs	NDK	measured	ratio
complete	320	353	1.10
expander	960	1,030	1.07
torus	1,280	1,319	1.03
ring	2,560	2,384	0.93
path	4,800	4,107	0.86

- Shape NDK matches the measured excess within $\pm 14\%$ on every graph (no fitted constant; the proved constant is larger).
- Almost flat in T, as predicted.
- Theorem 19 ($\Lambda \to \sqrt{\Lambda}$) is better when K is large.

Theorem 27: a provable relay, as good as the run one

The one-line change: every agent acts only on *published* records, including its own

Then a reward cannot change *when* its pull is published (Lemma 24), and anything it influences reaches every agent later (triangle inequality). So each view is **one prefix of a fixed order**, a martingale (Lemma 25): no $\sqrt{N}$, no epochs.

Theorem 27 ($N\gamma \le 1/2$, w.p. $\ge 1 - 4\delta$)

$$R_T(N) = \tilde{O}\Big((d\,NT)^{2/3} + K\,F'\Big), \qquad F' = \max_i \sum_l \max(d(i,l),1) \;\; \text{(farness)} \le ND$$

The graph enters through the **sum of distances**, not $4N\Lambda$; horizon-free. Star $F' \approx 2N$, ring $F' \approx ND/2$.

Measured: the provable relay is better than the one we ran

Excess over the server at the proved $\gamma = 1/(2N)$: complete -25 (vs 116), ring 381 (vs 528), path 488 (vs 633), for 7–8$\times$ more scalars. At $\gamma = 0.5$ it equals the run relay.

Experimental setup

Instances (as in Dey, Bhore & Ghosh)

$d = 10$, $K = 20$ contexts $\sim \mathcal{N}(0, I)$, noise $\sigma = 0.1$. Links: quadratic $-(z-1)^2 + 1$ (non-monotone), zigzag $\sin z + 0.3z$, logistic (monotone).

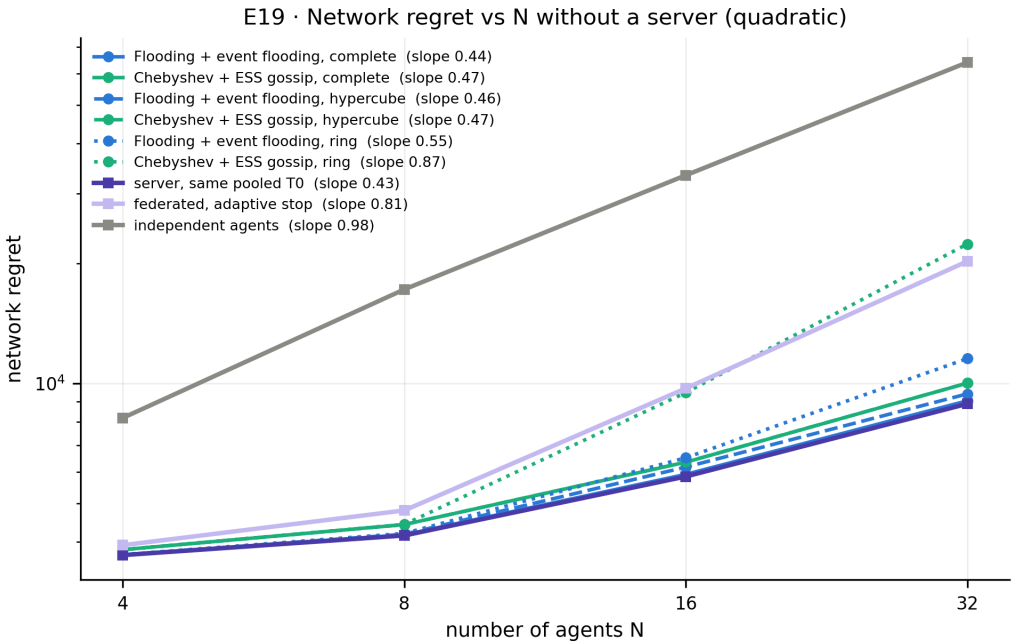
Networks

$N = 4$–32 agents (16 by default) on complete, expander, hypercube, torus, ring, path graphs ($D = 1$–15), plus failing, delayed, time-varying and directed links.

Protocol

- $T \leq 20{,}000$ per agent ($NT \leq 320$k); pooled Phase 1 $= 960\,(NT/160\text{k})^{2/3}$; $\Delta = (NT)^{-1/3}$; $\gamma = 0.5$ unless stated.
- 10–20 **paired** trials (the same seeds for every method), mean $\pm$ 95% CI.
- Metrics: network regret, *excess* over the server, scalars sent.
- References: server, independent agents. Baselines: consensus / gossip / push-sum protocols; prior algorithms with ideal relaying.

Results: collaboration pays without a server



quadratic link, fixed pooled Phase-1 size

Network regret $\propto N^{\alpha}$, $N = 4 \ldots 32$:

	α
server	0.43
flooding, complete	0.44
flooding, hypercube ($D = \log N$)	0.46
flooding, ring ($D = N/2$)	0.55
independent agents	0.98

At $N = 32$: hypercube within **6%** of the server, ring within **30%**.

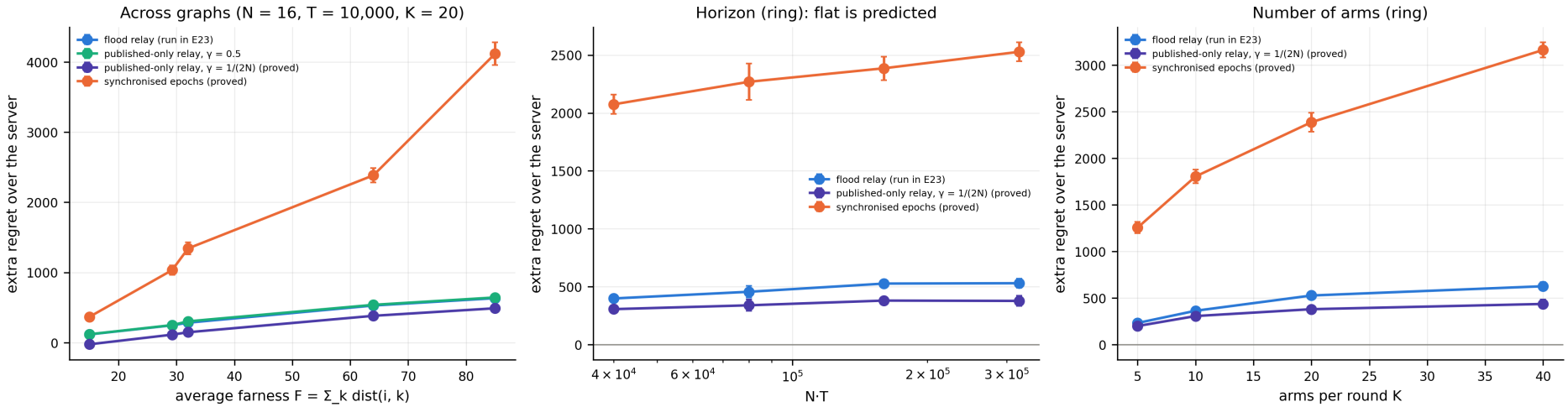
Results: relaying matches mixing with far fewer scalars

Design (torus, $N = 16$, same T_0)	**Regret ÷ server**	**Scalars sent**
Server + event-triggered sync	1.000	57k
Spanning tree + event-triggered flooding	1.030	60k
Flooding / consensus / push-sum every round	1.02	9.5M–295M

Link failures (ratio to server)	$q = 0.3$	$q = 0.6$	$q = 0.9$
Flooding + event flooding	1.06	1.09	1.27
Running consensus	1.04	1.10	1.50
Chebyshev (accelerated) gossip	2.1	3.5	7.3

- **Same regret, about 2000× fewer scalars:** consensus/push-sum need 140–285M scalars.
- On regret *alone*, mixing is not worse: consensus every round is as good or slightly better on well-connected graphs.
- Relaying depends on the *diameter*; mixing on the *spectral gap*, and acceleration collapses when links fail.

Results: the graph's cost has the proved shape



- **Across graphs:** epochs follow NDK (ratio 0.86–1.15); the relays follow the farness F (flood relay: excess$/KF = 0.37$–0.44 on every graph).
- **Horizon:** $8\times$ longer T raises the excess only $\times 1.2$–1.3, as Theorems 22 and 27 say (horizon-free up to logs).
- **Arms:** the excess grows like $K^{0.4}$–$K^{0.5}$, slower than the bounds' linear K. That factor is loose (open).

$N = 16$, quadratic link, 10 paired trials. The server reference uses $\gamma = 0.5$, so the proved relay ($\gamma = 1/32$) can sit slightly below it.

Theorem 29: communication

Theorem

Under flooding each message crosses each edge at most once. So the number of scalars is

$$\underbrace{2|E|N(d{+}1)}_{\text{Phase 1}} + \underbrace{2|E|\,|\mathcal{E}|\,\big(1 + N(3N_b{+}1)\big)}_{\text{Phase 2}}, \qquad |\mathcal{E}| \le 1 + N_b\big(1 + \log_{1+\gamma}(NT)\big).$$

- **Number of epochs:** between two alarms for a bin, its count grows by a factor $> 1+\gamma \Rightarrow$ logarithmically many epochs.
- For a fixed grid this is **logarithmic in** T, against $\Theta(|E|N_bT)$ for per-round sharing.

An honest trade-off

At the rate-optimal grid, $N_b \sim (NT)^{1/3}$, and with $\gamma = \gamma_0/N$ this becomes $\tilde{O}(|E|N^2(NT)^{2/3})$: polynomial in T. A constant γ needs $N\times$ fewer epochs, but the bound pays $\sqrt{1+N\gamma}$. In experiments $\gamma \le 0.5$ costs nothing; only $\gamma = 2$ hurts.

Robustness: only Λ changes (Lemma 31, Cor 32)

Network	**Delivery time Λ**
static, connected	D
directed, strongly connected	directed D
link delay $\leq \tau$	τD
union of every B rounds connected	$B(N-1)$, **tight**
subgraph stable for B rounds	D_B (its diameter)
links fail w.p. q	$\approx (D + \log NT)/(1-q)$

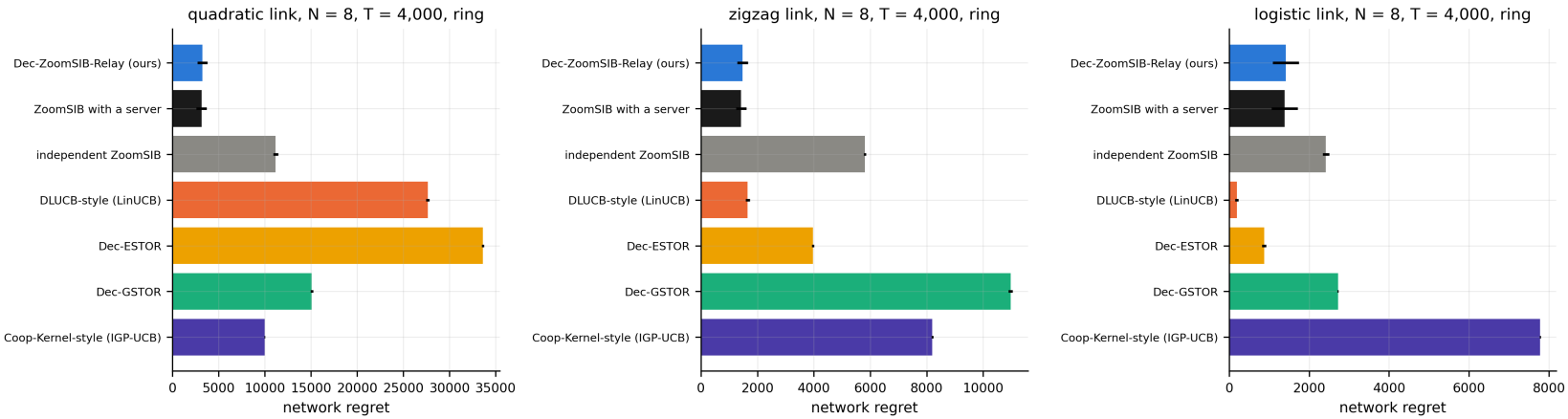
Why it is enough

The proofs use the graph only through Λ. Records arrive *late*, never *wrong*: the $(NT)^{2/3}$ term never changes, only the additive $N\Lambda(NT)^{1/3}$.

The diameter of the union graphs does not help

Each window, the only edges form a star centred at a new agent, all at the window's last round. Every union graph has diameter 2, yet only one new agent is informed per window $\Rightarrow \Lambda = B(N-1)$.

Results: prior papers' algorithms on our problem



- **Non-monotone link (quadratic):** ours is best, and the gap grows with T. Kernel UCB: $1.24\times$ ours at $T{=}1000$, $3.1\times$ at $T{=}4000$. Linear / monotone methods: 8.5–$10\times$.
- **Communication:** 7–70$\times$ fewer scalars than relaying raw data.
- **Monotone link (logistic):** LinUCB $(0.13\times)$ and ESTOR $(0.62\times)$ win. That is the price of not assuming monotonicity.
- **Mostly increasing (zigzag $= \sin z + 0.3z$):** LinUCB $0.54\times$ at $T{=}1000$, $1.12\times$ at $T{=}4000$.

$N = 8$, ring, $d = 10$, $K = 20$, 10 paired trials. Baselines get *ideal* relaying (every observation, exact, at graph-distance delay), at least as much information as their own gossip.

Theorem 33: agents must agree on the direction

Theorem

If agent k bins with its own $\hat{\theta}_k$ and $\max_k \|\hat{\theta}_k - \hat{\theta}\|_1 \le \eta_{\text{agree}}$ ($\hat{\theta}$ = exact pooled), the regret grows by at most

$$4L_{f'} X_{\max} \eta_{\text{agree}} NT.$$

- **Why:** directions enter the proof only through the projection error. Pooling a bin across agents still pools pulls whose means lie in the same interval.
- **Linear in NT:** even small disagreement hurts. It matters once $X_{\max}\eta_{\text{agree}}NT \sim (NT)^{2/3}$.
- **Fix:** one extra flood of the local Stein sums gives every agent the exact direction ($\eta_{\text{agree}} = 0$), for Λ rounds and $2|E|N(d{+}1)$ scalars.

Measured

Consensus Phase 1: $\times 1.25$ (ring), $\times 1.34$ (path). One flooding pass: $\times 1.08$, $\times 1.09$.

Conclusion and future work

Conclusion

- Without a server, agents learn an unknown index and link at the server's rate, plus an **additive graph cost that does not grow with T.**
- That cost is set by distances (farness); the measured cost has the proved shape.
- Relaying exact, sparse statistics: the regret of consensus, orders of magnitude fewer scalars, robust to failing links.

Limitations

- Graph lower bound $N \min(T, D)^{1/3}$ vs upper KF': a gap. Bounds linear in K; we measure $\approx \sqrt{K}$.
- Synthetic data only; private Phase 1 is costly; private Phase 2 is open. The newest proofs are still being checked.

Future work

Close the graph gap; gap-dependent K; private Phase 2; heterogeneous (θ_i^*) and asynchronous agents.

References

1. D. Dey, S. Bhore, A. Ghosh. *Optimal Regret for Single Index Bandits.* arXiv:2605.09454, 2026.
2. Y. Kang et al. *Single Index Bandits: Generalized Linear Contextual Bandits with Unknown Reward Functions.* ICLR, 2026.
3. A. Dubey, A. Pentland. *Kernel Methods for Cooperative Multi-Agent Contextual Bandits.* ICML, 2020.
4. S. Amani, C. Thrampoulidis. *Decentralized Multi-Agent Linear Bandits with Safety Constraints.* AAAI, 2021.
5. D. Martínez-Rubio, V. Kanade, P. Rebeschini. *Decentralized Cooperative Stochastic Bandits.* NeurIPS, 2019.
6. P. Landgren, V. Srivastava, N. E. Leonard. *On Distributed Cooperative Decision-Making in Multiarmed Bandits.* ECC, 2016.
7. Y. Wang, J. Hu, X. Chen, L. Wang. *Distributed Bandit Learning: Near-Optimal Regret with Efficient Communication.* ICLR, 2020.
8. S. Salgia, Q. Zhao. *Distributed Linear Bandits under Communication Constraints.* arXiv:2211.02212, 2022.
9. T. Lattimore, C. Szepesvári. *Bandit Algorithms.* Cambridge University Press, 2020.

Thank You!